

Major Assignment – 4 (Minor Project)

1. Spam Email Classification using Naïve Bayes

Objective	To classify emails as <i>Spam</i> or <i>Not Spam</i> using probabilistic machine learning.
Problem Description	With the rapid growth of online communication, spam emails create security and productivity issues. An automated spam detection system is required.
Dataset	- Email text messages (https://www.kaggle.com/datasets/tmehul/spamcsv) - Labels: Spam / Ham
Methodology	- Text preprocessing (tokenization, stop-word removal) - Feature extraction using word frequencies - Apply Naïve Bayes theorem - Calculate posterior probabilities - Classify emails using MAP decision rule
Algorithm Used	Naïve Bayes Classifier
Tools	- Python - Scikit-learn
Expected Outcome	Accurate identification of spam emails with high efficiency.

2. Disease Diagnosis using Support Vector Machine (SVM)

Objective	To diagnose diseases using patient medical data obtained from a standard machine learning dataset.
Problem Description	Manual diagnosis of diseases can be prone to error and time-consuming. Automated decision support systems that utilise machine learning techniques enhance diagnostic reliability and consistency.
Dataset	The study uses the Diabetes dataset from <code>sklearn.datasets</code> , which contains real-valued medical features such as: - Age - Sex - Body Mass Index (BMI) - Blood pressure - Serum measurements (six attributes)
Methodology	- Load the diabetes dataset using <code>sklearn.datasets</code> - Perform data normalization and preprocessing - Identify the optimal separating hyperplane - Maximize the margin between classes - Apply kernel-based transformation using RBF kernel - Classify disease outcomes using SVM
Algorithm Used	Support Vector Machine (SVM)
Tools	- Python - Scikit-learn
Expected Outcome	Accurate disease prediction using the diabetes dataset. (Evaluation report: Accuracy, Precision, Recall, F1-score and Confusion matrix)

3. Credit Card Fraud Detection using Random Forest

Objective	To detect fraudulent credit card transactions.
Problem Description	Online transactions are vulnerable to fraud, causing financial losses.
Dataset	From Kaggle: https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud Place creditcard.csv in your working directory
Methodology	• Data cleaning • Feature selection • Build multiple decision trees • Aggregate predictions using voting • Evaluate performance
Algorithm Used	Random Forest Classifier
Tools	• Python • Scikit-learn
Expected Outcome	High fraud detection accuracy with reduced false alarms. (Evaluation report: Accuracy, Precision, Recall, F1-score and Confusion matrix)

4. Student Performance Prediction using Decision Tree

Objective
To predict student academic performance using a Decision Tree model based on demographic and academic attributes.
Problem Description
Educational institutions aim to identify students at risk of poor performance. Decision Tree models provide interpretable rules that help understand the factors influencing student outcomes.
Dataset
Student Performance Dataset (e.g., from UCI Machine Learning Repository), including: • Study time • Attendance • Previous grades • Parental education • Internet access
Methodology
• Load and preprocess the dataset • Handle missing values and categorical variables • Split data into training and testing sets • Train a Decision Tree classifier • Visualize the decision tree structure • Prune the tree to avoid overfitting • Evaluate model performance
Algorithm Used
Decision Tree Classifier
Tools

• Python • Scikit-learn • Matplotlib
Expected Outcome
Clear and interpretable classification rules with good prediction accuracy, evaluated using Accuracy, Precision, Recall, F1-score, and Confusion Matrix.

5. House Price Prediction using Linear Regression

Objective
To predict house prices based on various housing features using a Linear Regression model.
Problem Description
Real-estate pricing depends on multiple factors such as location, size, and number of rooms. Linear regression provides a simple yet effective method for modeling the relationship between house prices and influencing variables.
Dataset
Boston Housing dataset (https://www.kaggle.com) containing features such as: • Number of rooms • Crime rate • Distance to employment centers • Property tax rate • Median house value
Methodology
• Load the dataset • Perform exploratory data analysis • Handle multicollinearity and normalize features • Split the data into training and testing sets • Train a Linear Regression model • Predict house prices on test data • Evaluate regression performance
Algorithm Used
Linear Regression
Tools
• Python • Scikit-learn • NumPy • Pandas
Expected Outcome
Accurate house price predictions evaluated using Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R² score.